1. Vectors & Basic Ops

$e_i^T a = a_i$
Inner product: $x^T y = \sum x_i y_i$, $\approx 2n$ flops.
Norms: $\|x\|_2 = \sqrt{x^T x}$, $\|x\|_1 = \sum |x_i|$, $\|x\|_\infty = \max |x_i|$
Angle: $x^T y = \|x\|\|y\| \cos\theta$
Cauchy-Schwarz: $|x^T y| \le \|x\|\|y\|$
Linear independence: $\sum \alpha_i v_i = 0 \Rightarrow \alpha_i = 0$
Basis: n independent vectors in $\mathbb{R}^n$
Orthonormal: $v_i^T v_j = 0$ for $i \ne j$, $\|v_i\| = 1$; n orthonormal vectors in $\mathbb{R}^n$ form ON basis.
Affine function: $f(x) = a^T x + b$ (linear + constant).

2. Matrices & Linear Systems

Positive Definite: $x^T A x > 0\ \forall x \ne 0$.
Positive Semidefinite: $x^T A x \ge 0\ \forall x$;
A^{-1} exists iff rows/cols lin. indep. iff rows/cols form basis
Solve $Ax = b$: Diagonal $O(n)$, Triangular $O(n^2)$, Permutation $O(0)$
Solving Linear Systems:
LU: $A = PLU$, factor $\frac{2}{3}n^3$, solve $2n^2$.
Cholesky: $A = LL^T$ for symmetric PD, factor $\frac{1}{3}n^3$, solve $2n^2$.

3. Least Squares (L3)

Tall $A \in \mathbb{R}^{m\times n}$, $m > n$: usually no exact $Ax = b$.
LS problem: $\min_x \|Ax - b\|_2^2$, residual $r = Ax - b$.
Column view: LS finds $Ax^* \in \text{col}(A)$ closest to b
Gradient: $\nabla \|Ax - b\|_2^2 = 2A^T(Ax - b)$.
Optimality: $A^T(Ax^* - b) = 0 \Rightarrow$ residual $r^* \perp \text{col}(A)$.
Solve: $A^T A x^* = A^T b$.
If A full col rank: $x^* = (A^T A)^{-1} A^T b$
Gram $G = A^T A$: symmetric PSD always; PD/invertible iff A has indep. cols.
Solve via Cholesky: factor $A^T A = LL^T$, solve $LL^T x = A^T b$
Complexity: form $A^T A$: $2mn^2$; Cholesky: $\frac{1}{3}n^3$; prefactored solve (new b): $\approx 2mn$

4. Least Squares Data Fitting

Data: $(x^{(i)}, y^{(i)})$, $i = 1, \dots, N$.
Model: $\hat{f}(x) = \theta_1 f_1(x) + \cdots + \theta_p f_p(x)$, linear in θ.
Matrix form: $A_{ij} = f_j(x^{(i)})$, $\hat{y} = A\theta$.
LS: $\min_\theta \|A\theta - y\|_2^2$, $A^T A \theta^* = A^T y$.
Residual $r^{(i)} = y^{(i)} - \hat{y}^{(i)}$. MSE $= \frac{1}{N}\sum_i (r^{(i)})^2 = \frac{1}{N}\|A\theta - y\|_2^2$; RMSE $= \sqrt{\text{MSE}}$.
Line: $\hat{f}(x) = \theta_1 + \theta_2 x$, $A = [\mathbf{1}\ \ x_d]$.
Polynomial: $f_j(x) = x^{j-1}$, $\hat{f}(x) = \sum_{j=1}^p \theta_j x^{j-1}$, $A_{ij} = (x^{(i)})^{j-1}$ (Vandermonde).
AR(M): $\hat{z}_{t+1} = \theta_1 z_t + \cdots + \theta_M z_{t-M+1}$; $y^{(i)} = z_{M+i}$, $x^{(i)} = (z_{M+i-1}, \dots, z_i)$.
General data fitting = linear regression on transformed features: $\tilde{x} = (f_2(x), \dots, f_p(x))$, intercept $f_1(x) = 1$.
Validation: train/test split; overfitting = low train MSE + high test MSE. k-fold CV: train on $k-1$ folds, test on held-out fold; choose lowest test error, then retrain on all data.

5. Multi-Objective Least squares

Problem: $\min_x \sum_{i=1}^k \lambda_i \|A_i x - b_i\|^2$, with $\lambda_i > 0$.

Equivalent stacked LS: $\min_x \left\| \begin{bmatrix} \sqrt{\lambda_1} A_1 \\ \vdots \\ \sqrt{\lambda_k} A_k \end{bmatrix} x - \begin{bmatrix} \sqrt{\lambda_1} b_1 \\ \vdots \\ \sqrt{\lambda_k} b_k \end{bmatrix} \right\|^2$

Solution: $(\tilde{A}^T \tilde{A}) x^* = \tilde{A}^T \tilde{b}$

Sum of distances to points: $\sum_{i=1}^k \|x - a_i\|^2 = \left\| \begin{bmatrix} I \\ \vdots \\ I \end{bmatrix} x - \begin{bmatrix} a_1 \\ \vdots \\ a_k \end{bmatrix} \right\|^2$.

Here $A \in \mathbb{R}^{kn\times n}$ stacks k copies of I_n, $b \in \mathbb{R}^{kn}$ stacks a_i.
Bi-criterion: $\min J_1 + \lambda J_2$. If J_2 too big, increase λ; if J_1 too big, decrease λ.
Trade-off curve: $(J_1(x^*(\lambda)), J_2(x^*(\lambda)))$, Pareto optimal
Tikhonov: $\min \|Ax - y\|^2 + \lambda\|x\|^2 = \left\| \begin{bmatrix} A \\ \sqrt{\lambda} I \end{bmatrix} x - \begin{bmatrix} y \\ 0 \end{bmatrix} \right\|^2$
Normal equation: $(A^T A + \lambda I)x = A^T y$
Smoothing: $\min \|Ax - y\|^2 + \lambda\|Dx\|^2$ (D = diff matrix)
Normal equation: $(A^T A + \lambda D^T D)x = A^T y$

6. Constrained Least Squares

Problem: $\min \|Ax - b\|^2$ subject to $Cx = d$
Lagrangian $L(x, z) = \|Ax - b\|^2 + z^T(Cx - d)$.
KKT: $\begin{bmatrix} 2A^T A & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} 2A^T b \\ d \end{bmatrix}$
KKT invertible iff C has indep. rows and $[A; C]$ has indep cols
Least norm: $\min_x \|x\|^2$ s.t. $Cx = d$ = CLS with $A = I, b = 0$.
Portfolio: weights w, $\mathbf{1}^T w = 1$; $w_i < 0$ short; leverage $L = \|w\|_1$.
Return matrix $R \in \mathbb{R}^{T\times n}$, portfolio returns $r = Rw$. $\text{avg}(r) = \mathbf{1}^T r / T$, $\text{std}(r) = \|r - \text{avg}(r)\mathbf{1}\| / \sqrt{T}$.
Annualized return $= P\,\text{avg}(r)$; annualized risk $= \sqrt{P}\,\text{std}(r)$.
Portfolio CLS: $\min_w \|Rw - \rho\mathbf{1}\|^2$ s.t. $\mathbf{1}^T w = 1$, $\mu^T w = \rho$, $\mu = R^T \mathbf{1} / T$.
KKT: $\begin{bmatrix} 2R^T R & \mathbf{1} & \mu \\ \mathbf{1}^T & 0 & 0 \\ \mu^T & 0 & 0 \end{bmatrix} \begin{bmatrix} w \\ z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 2\rho T\mu \\ 1 \\ \rho \end{bmatrix}$.
Two-fund theorem: $w^*(\rho) = w_0 + \rho v$ affine in target return.

7. Linear Optimization / LP

LP general form: $\min c^T x$ subj. to $Ax \le b$, $Dx = f$; inequalities are computed elementwise.
Feasible x satisfies constraints. x^* optimal if feasible and $c^T x^* \le c^T x$ for all feasible x. Optimal value $p^* = c^T x^*$.
Special cases: infeasible $\Rightarrow p^* = +\infty$; unbounded below $\Rightarrow p^* = -\infty$; optimizer may be nonunique.
Feasibility problem (find any x subj. to LP constraints): $\min 0$ subj. to constraints of the LP
Standard form: $\min c^T x$ s.t. $Ax = b$, $x \ge 0$.
Transformations to standard form:
maximize $c^T x \Rightarrow$ minimize $-c^T x$.
$Ax \le b \Rightarrow Ax + s = b,\ s \ge 0$;
$Ax \ge b \Rightarrow Ax - s = b,\ s \ge 0$.
$x_i \le 0 \Rightarrow x_i = -y_i,\ y_i \ge 0$;
x_i free $\Rightarrow x_i = x_i^+ - x_i^-$, $x_i^+, x_i^- \ge 0$.

8. Piecewise Linear Optimization

Convex PL: $f(x) = \max_i (a_i^T x + b_i)$
Minimization: $\min_x f(x) \equiv \min t$ s.t. $a_i^T x + b_i \le t,\ \forall i$.
Sum of PL: use separate t_j for each max term; minimize $\sum_j t_j$.
$\min \|Ax - b\|_\infty$: $\min t$ s.t. $-t\mathbf{1} \le Ax - b \le t\mathbf{1}$
$\min \|Ax - b\|_1$: $\min \mathbf{1}^T u$ s.t. $-u \le Ax - b \le u$
ℓ_2^2: sum squared errors. ℓ_1: sum absolute errors $\sum_i |r_i|$. ℓ_∞: max absolute error $\max_i |r_i|$.
General abs trick: $|y_i| \le u_i \iff y_i \le u_i,\ -y_i \le u_i$.
Constraint $\|x\|_1 \le k$: introduce u, $-u \le x \le u$, $\mathbf{1}^T u \le k$.
Sparse recovery: $\min_x \|x\|_1$ s.t. $Ax = y \equiv \min_{x,u} \mathbf{1}^T u$ s.t. $-u \le x \le u$, $Ax = y$; ℓ_1 promotes sparsity.
SVM separable: find a, b s.t. $s_i(a^T v_i + b) \ge 1$, $s_i \in \{-1, 1\}$; LP feasibility.
Nonseparable hinge loss: $\min_{a,b} \sum_i \max\{0, 1 - s_i(a^T v_i + b)\} \equiv \min \mathbf{1}^T u$ s.t. $u_i \ge 0$, $u_i \ge 1 - s_i(a^T v_i + b)$.
Worst-case max of linear costs: $\min_x \max_k (p^k)^T x \equiv \min_{x,v} v$ s.t. $(p^k)^T x \le v,\ \forall k$.
Smoothness / total variation: $\sum_{t=1}^{T-1} |x_{t+1} - x_t|$
LP via $y_t \ge x_{t+1} - x_t$, $y_t \ge -(x_{t+1} - x_t)$; minimize $\mathbf{1}^T y$.

9. Geometry & Polyhedra (L9)

Hyperplane: $\{x \mid a^T x = b\}$; halfspace: $\{x \mid a^T x \le b\}$. Polyhedron: $P = \{x \mid Ax \le b\}$ = finite halfspaces
Extreme point: cannot write $x = \alpha y + (1 - \alpha) z$ with $y, z \in P$, $y, z \ne x$, $\alpha \in (0, 1)$.
Vertex: unique minimizer of some linear objective over P.
Active constraints: $I(\bar{x}) = \{i \mid a_i^T \bar{x} = b_i\}$.
Standard form $P = \{x \mid Ax = b,\ x \ge 0\}$, $A \in \mathbb{R}^{m\times n}$ full row rank.
Choose m indep. columns A_B, set $x_N = 0$, solve $A_B x_B = b$. If $x_B \ge 0$, then x is a Basic Feasible Sol (BFS).
Equivalence: vertex $\iff$ extreme point $\iff$ BFS.
Degenerate BFS: $> n$ active constraints (in standard form, some basic variable $= 0$)
P has an extreme point iff P contains no line. Nonempty bounded polyhedron has a BFS. If LP optimum exists and P has an extreme point, then some optimum is an extreme point.

11. The Simplex Method

Idea: move BFS $\to$ neighboring BFS with lower cost
Reduced cost: $\bar{c}_j = c_j - c_B^T A_B^{-1} A_j$; $\bar{c}_j = 0$ for basic
Optimal: all $\bar{c}_j \ge 0$
If $\bar{c}_j < 0$, bringing x_j into basis decreases cost
Feasible direction: $d_j = 1$, solve $A_B d_B = -A_j$
Step length: $\theta^* = \min_{\{i \in B \mid d_i < 0\}} (-x_i / d_i)$ (exiting index)
Unbounded: if $d_B \ge 0$, then $x + \theta d$ feasible $\forall\theta \ge 0$ and cost $\to -\infty$.
Simplex iteration: (1) $A_B^T p = c_B$, $\bar{c} = c - A^T p$; (2) if $\bar{c} \ge 0$ stop; (3) choose j with $\bar{c}_j < 0$; (4) $A_B d_B = -A_j$, $d_j = 1$; (5) if $d_B \ge 0$ unbounded; (6) θ^* from min ratio; (7) $x \leftarrow x + \theta^* d$; (8) update basis
Finite convergence (if non-degenerate): objective strictly decreases cost and there are finitely many BFSs
Degeneracy can cause cycling; Bland's rule (smallest index entering/leaving) prevents cycling

12. Simplex Implementation

Need initial BFS for $\min c^T x$ s.t. $Ax = b$, $x \geq 0$.
Phase (I): assume $b \geq 0$ wlog; if $b_i < 0$, multiply row i by -1. add artificial y, $\min \mathbf{1}^T y$ s.t. $Ax + y = b$, $x \geq 0, y \geq 0$, start $x = 0, y = b$;
if opt > 0 original LP infeasible; if $= 0$ get BFS, drop y, goto (II)
Phase (II): solve original from Phase I basis
Bland's rule: entering = smallest index with $\bar{c}_j < 0$; leaving = smallest index attaining min ratio
Complexity: per iteration $O(n^2)$ (matrix update); worst-case 2^n; average $O(n)$

13–14. Duality

Standard primal: $\min c^T x$ s.t. $Ax = b$, $x \geq 0 \Rightarrow$ dual: $\max -b^T y$ s.t. $A^T y + c \geq 0$, y free.
Inequality primal: $\min c^T x$ s.t. $Ax \leq b \Rightarrow$ dual: $\max -b^T y$ s.t. $A^T y + c = 0$, $y \geq 0$.
Mixed primal: $\min c^T x$ s.t. $Ax \leq b$, $Cx = d \Rightarrow$ dual: $\max -b^T y - d^T z$ s.t. $A^T y + C^T z + c = 0$, $y \geq 0$, z free.
Sign convention: write constraints as $Ax - b \leq 0$ or $Ax - b = 0$. Inequality dual vars ≥ 0; equality dual vars free.
Weak duality: primal feasible x, dual feasible $y \Rightarrow$ dual obj $\leq$ primal obj; for inequality form $-b^T y \leq c^T x$.
Duality gap: $c^T x + b^T y \geq 0$; if gap $= 0$, x, y optimal.
Strong duality: if primal has optimum, dual has optimum and $p^* = d^*$. Primal unbounded $\Rightarrow$ dual infeasible; dual unbounded $\Rightarrow$ primal infeasible.
Complementary slackness: $y_i(b_i - a_i^T x) = 0$. Hence $y_i > 0 \Rightarrow a_i^T x = b_i$; $a_i^T x < b_i \Rightarrow y_i = 0$.
KKT (checklist for proving a sol is opt.): primal feas ($Ax \leq b$); dual feas ($A^T y + c = 0$, $y \geq 0$); comp. slack ($y_i(Ax - b)_i = 0\ \forall i$)
Farkas: exactly one holds: (i) $\exists x \geq 0$ s.t. $Ax = b$; (ii) $\exists y$ s.t. $A^T y \geq 0$, $b^T y < 0$.

15. Sensitivity Analysis

Add variable x_{n+1}: old $(x^*, 0)$ still primal feasible. still opt iff $A_{n+1}^T y^* + c_{n+1} \geq 0$; else add variable & use primal simplex
Add constraint $a_{m+1}^T x = b_{m+1}$: old dual $(y^*, 0)$ still dual feasible. still optimal iff $a_{m+1}^T x^* = b_{m+1}$; else add constraint & use dual simplex.
RHS perturbation: $p^*(u) = \min\{c^T x \mid Ax = b + u,\ x \geq 0\}$ = optimal val when RHS b is changed to $b + u$. Global lower bound: $p^*(u) \geq p^*(0) - u^T y^*$.
If basis B unchanged:

- $x_B^*(u) = A_B^{-1}(b + u) = x_B^* + A_B^{-1} u$, $x_N^*(u) = 0$
- $p^*(u) = p^*(0) - u^T y^* = -(b + u)^T y^*$
- $\nabla p^*(0) = -y^*$ = shadow price; one more unit of RHS i changes objective by $-y_i^*$
- Objective perturbation: if $c \to c + \epsilon d$ and basis unchanged, $p^*(\epsilon) = p^*(0) + \epsilon d^T x^*$.

16. Network Flow

Directed network: m nodes, n arcs
Arc-node incidence $A \in \mathbb{R}^{m \times n}$: $A_{ij} = 1$ if arc j leaves i, -1 if enters, else 0
Min cost flow: $\min c^T x$ s.t. $Ax = b$, $0 \leq x \leq u$, $\mathbf{1}^T b = 0$
Total unimodularity: node-arc incidence is TU
Integrality: if A TU and b, u integer, BFS are integer
Max flow: add arc sink$\to$source with cost -1, cap ∞; source 1, sink m, $e = (1, 0, \ldots, 0, -1)$: $\max t$ s.t. $Ax = te$, $0 \leq x \leq u$.
As min-cost flow: $\min -t$ s.t. $[A\ \ -e]\binom{x}{t} = 0$, $0 \leq x \leq u$, $t \geq 0$.
Shortest path from 1 to m: $\min c^T x$ s.t. $Ax = e$, $0 \leq x \leq 1$, where $e = (1, 0, \ldots, 0, -1)$
By TU, LP solution is binary, so $x_j = 1$ iff arc j is in path.
Assignment: nodes = people & tasks; arcs = possible assignments. x_k = flow on arc k ($x_{ij} = 1$ if person i assigned to task j, else 0). $A \in \mathbb{R}^{(\#nodes) \times (\#arcs)}$ incidence matrix: $A_{ik} = 1$ if arc k leaves node i, -1 if arc k enters node i, 0 otherwise. $b = (\mathbf{1}_N, -\mathbf{1}_N)$: people have supply $+1$, tasks have demand -1. Solve $\min c^T x$ s.t. $Ax = b$, $0 \leq x \leq 1$.

17–18. Interior-Point Methods

Newton: solve $h(x) = 0$, $x^{k+1} = x^k + \Delta x$, $Dh(x^k)\Delta x = -h(x^k)$
For $\min c^T x$ s.t. $Ax \leq b$, add slack $s \geq 0$: $Ax + s = b$.
LP optimality: $h = [Ax + s - b;\ A^T y + c;\ SY\mathbf{1}] = 0$, $s, y \geq 0$
Residuals: $r_p = Ax + s - b$, $r_d = A^T y + c$, $\mu = s^T y / m$
Central path (smoothed): $s_i y_i = \tau$, $s, y > 0$; follow as $\tau \to 0$
Newton system ($\sigma \in [0, 1]$, $\sigma = 1$ centering, $\sigma = 0$ pure Newton):

$$\begin{bmatrix} 0 & A & I \\ A^T & 0 & 0 \\ S & 0 & Y \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta x \\ \Delta s \end{bmatrix} = \begin{bmatrix} -r_p \\ -r_d \\ -SY\mathbf{1} + \sigma\mu\mathbf{1} \end{bmatrix}$$

Primal-dual path-following: choose σ, solve system, line search for max α keeping $s, y > 0$, update.
Barrier: $\min c^T x - \tau \sum_i \log(s_i)$ s.t. $Ax + s = b$; barrier keeps $s_i > 0$.
Stop when $\|r_p\|, \|r_d\|$, $s^T y$ are small.
Mehrotra predictor-corrector:
1. Predictor ($\sigma = 0$), get Δ_a, compute $\alpha_p, \alpha_d, \mu_a$
2. $\sigma = (\mu_a/\mu)^3$
3. Corrector RHS $-SY\mathbf{1} - \Delta S_a \Delta Y_a \mathbf{1} + \sigma\mu\mathbf{1}$

Complexity: IPM average $O(\log n)$ iterations, $O(n^3)$ per iter
IPM vs Simplex: IPM variants polynomial, hard to warm-start, good large/sparse; Simplex exponential worst-case, needs BFS/Phase I, can warm-start, good small/medium.

20–21. Integer Optimization

Problem: $\min c^T x$ s.t. $Ax \leq b$, $x_i \in \mathbb{Z}$ for $i \in I$
Binary: $x_i \in \{0, 1\}$. LP relaxation removes integrality; for min, $p_{\text{rel}}^* \leq p_{\text{IP}}^*$.
Logic: at most one $\mathbf{1}^T x \leq 1$; x_1 only if x_2: $x_1 \leq x_2$; both/neither: $x_1 = x_2$.
Knapsack: $\max c^T x$ s.t. $a^T x \leq b$, $x_i \in \{0, 1\}$.
If-then ($y = 0 \Rightarrow x = 0$) via $0 \leq x \leq My$ (Big-M)
Disjunction: $a^T x \leq b$ or $d^T x \leq f$ via $a^T x \leq b + yM$, $d^T x \leq f + (1 - y)M$.
Cardinality: $\text{card}(x) \leq k$ via $\sum_i y_i \leq k$, $-My_i \leq x_i \leq My_i$, $y_i \in \{0, 1\}$.
Restricted values $x \in \{a_1, \ldots, a_d\}$: $x = a^T z$, $\mathbf{1}^T z = 1$, $z \in \{0, 1\}^d$.
Better formulation = tighter relaxation. If $P_{\text{rel},1} \subset P_{\text{rel},2}$, then $p_{\text{rel},2}^* \leq p_{\text{rel},1}^* \leq p_{\text{IP}}^*$ for min.
Ideal formulation: LP relaxation gives integer optimum
Branch-and-bound:
- Bound: L_j (relaxation lower bound), U_j (heuristic upper bound)
- Global: $L = \min L_j$, $U = \min U_j$, gap $U - L$
- Branch: if $\bar{x}_i$ fractional, split into $x_i \leq \lfloor \bar{x}_i \rfloor$ and $x_i \geq \lceil \bar{x}_i \rceil$.
- Prune if relaxation infeasible, relaxation solution integer, or $L_j \geq U$

Incumbent = best integer feasible solution found so far; its value gives global upper bound U

22–23. Robust Optimization / Uncertainty

Uncertain constraint: $a^T x \leq b$ with $a \in U$
Robust constr: $a^T x \leq b\ \forall a \in U \iff \max_{a \in U} a^T x \leq b$.
Parametrize $a = \bar{a} + Mu$, $u \in U$. Let $w = M^T x$. Then robust constraint is $\bar{a}^T x + \max_{u \in U} w^T u \leq b$.
Box uncertainty ($\|u\|_\infty \leq \rho$): $\max w^T u = \rho\|w\|_1$
Robust: $\bar{a}^T x + \rho\|M^T x\|_1 \leq b$
LP: $\bar{a}^T x + \rho\mathbf{1}^T t \leq b$, $-t \leq M^T x \leq t$.
1-norm uncertainty ($\|u\|_1 \leq \rho$): $\max w^T u = \rho\|w\|_\infty$
Robust: $\bar{a}^T x + \rho\|M^T x\|_\infty \leq b$
LP: $\bar{a}^T x + \rho\tau \leq b$, $-\tau \leq (M^T x)_i \leq \tau$.
Polyhedral uncertainty ($U = \{u \mid Fu \leq g\}$): $\max_{Fu \leq g} w^T u$ dualizes to $\min_\lambda g^T \lambda$ s.t. $F^T \lambda = w$, $\lambda \geq 0$
Robust LP: $\bar{a}^T x + g^T \lambda \leq b$, $F^T \lambda = M^T x$, $\lambda \geq 0$.
Larger ρ = more conservative. Robust solution trades nominal objective for better worst-case feasibility/performance.

Scenario Approach:
Given samples $d_1, \ldots, d_N$ of uncertain coefficient a, replace robust constraint $a^T x \leq b\ \forall a \in U$ with sample constraints $d_i^T x \leq b$, $i = 1, \ldots, N$.
Scenario LP: $\min c^T x$ s.t. $d_i^T x \leq b$, $i = 1, \ldots, N$. Dataset itself becomes uncertainty set.
Sample bound: if d_i i.i.d., choose $N \geq \frac{2}{\epsilon}\left(\ln\frac{1}{\beta} + n - 1\right)$ to get violation probability $\leq \epsilon$ with confidence $\geq 1 - \beta$.
n = decision variables, ϵ = violation tolerance, β = confidence failure probability. Pros: distribution-free, simple, no reformulation. Cons: random solution, may violate unseen scenarios, larger N increases solve time.